

23CSE452 Business Analytics · Individual Case Study

Predicting Used-Car Asking Prices in India: Age, Kilometres, Brand and Depreciation

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1. Problem Statement and Objectives

The Indian used-car market contains vehicles that differ widely in age, kilometres driven, fuel type, transmission, body type, ownership, brand and location. A dealer or private seller must combine all these factors into one asking price. If the price is too high, the vehicle remains unsold and blocks working capital; if it is too low, the seller loses margin. In practice, pricing is often based on one nearby advertisement or a fixed annual depreciation rule, even though two cars of the same model can have very different usage, ownership and market conditions.

This case study examines whether public listing attributes can provide a more defensible pricing method. The unit of analysis is one Cars24 advertisement—for example, a 2019 Swift with 40,000 kilometres in Pune. The target variable is the advertised asking price in lakh. It is important to distinguish this from the final negotiated sale price, which is not available on the public listing page.

The objectives of this case study are:

- To analyse the relationship between vehicle age, kilometres driven and used-car asking price, including residual value where a current new-car mid price can be matched.
- To examine the effect of vehicle characteristics such as brand, fuel type, transmission, body type, ownership and city on asking price.
- To develop and compare Linear Regression, Decision Tree and Random Forest for predicting used-car asking prices, and evaluate them on the same split using R^2 , MAE and RMSE.

2. Data Collection and Dataset Description

Used listings were collected from the same public Cars24 search pages in two ways. Both used a grid of 745 start URLs (city × brand × fuel × body type). Each page already embeds listing JSON, so year, kilometres, fuel, transmission, ownership and asking price are on every card. Packaged datasets from Kaggle, UCI or GitHub were not used.

The first method is a local Python scraper (requests). It reads the embedded JSON and keeps one row per appointment id while crawling. It returned 6,693 unique listings. After cleaning, these records form the analysis dataset used throughout the case study.

The second method is Apify Web Scraper, using the same start URLs and the same fields in the page function. The full run stored 12,560 rows because filter pages overlap and the actor did not drop duplicates while crawling; after unique appointment ids are kept, 6,766 listings remain. Figure 1 shows this completed run (12,560 results, \$1.597). The two collections return a similar number of unique cars

and the same field layout. They are not the same 6,700 listings, because stock can change between runs. Modelling uses the cleaned Python scrape only, so the two snapshots are not combined.

#	Appointment_id	Asking_price_lakh	Body_type	Brand	City	Fuel_type	Km_driven	Listing_price_inr	Locality	Model	Original_price_inr	Owner
1	16362876155	6.01	SUV	Tata	New Delhi	Petrol	52024	600936	Radisson Blu, Sector 1...	NEXON	640132	
2	11074178175	3.85	SUV	Nissan	new delhi	Petrol	46817	385320	Rajouri Garden	MAGNITE	387531	
3	15193399156	3.42	SUV	Tata	New Delhi	Petrol	94122	342252	Metro Walk, Rohini,...	NEXON	425079	
4	16324697135	2.67	SUV	Ford	New Delhi	Petrol	104834	267044	Radisson Blu, Sector 1...	Ecosport	339248	
5	10031195133	4	SUV	Maruti	New Delhi	Petrol	70432	400498	Metro Walk, Rohini,...	Ertiga	464500	

Figure 1. Apify Web Scraper run: 12,560 stored results (6,766 unique appointment ids after duplicates are removed).

Residual value uses a second, smaller public scrape of current new-car prices. Public CarDekho new-model pages were collected with Python (requests and BeautifulSoup). The reference covers 311 current nameplates across 44 brands, using the midpoint of each published ex-showroom range. It is a lookup only: it is not combined with the Cars24 listings for modelling, and new-car price is not a predictor. Used listings are matched on brand and model; unmatched cars keep a blank residual percent.

After cleaning, 6,629 passenger used cars remain, covering 22 brands, 212 models and 93 cities. Each row is one advertised car, with brand, model, variant, year, kilometres, fuel, transmission, body type, owner count, asking price, city, RTO and whether the car is inspected stock or a customer-to-customer listing. Roughly half the sample is each stock type. The modelling target is advertised asking price, not a completed sale. The crossed-out figure on some cards is the previous ask, not the first owner's invoice.

Dataset variables

The cleaned analysis dataset has 23 columns. The first 18 come from the Cars24 scrape (locality is dropped before submission to avoid exposing seller-entered addresses/phone numbers in customer-to-customer listings; it was never a model input). The last five are derived in the notebook: age, join keys for the new-car match, residual percent and the matched new mid-price. Residual percent and new mid-price are not model inputs.

Column	Type	Role
source_url	Text	Listing page URL
appointment_id	ID	Unique Cars24 listing id (dedupe key)
brand, model, variant	Categorical	Manufacturer, nameplate, variant

year	Numerical	Manufacturing year
km_driven	Numerical	Odometer (km)
fuel_type, transmission, body_type	Categorical	Fuel, gearbox, body style
ownership	Numerical	Number of previous owners
listing_price_inr	Numerical	Asking price in rupees
asking_price_lakh	Numerical	Same price in lakh (modelling target Y)
original_price_inr	Numerical	Previous/strikethrough ask on the card, not the first invoice
city, rto	Categorical	Location and RTO
stock_type	Categorical	Inspected stock or customer-to-customer
scraped_at	Datetime	When the Python scrape stored the row
age_years	Numerical	Derived: 2026 – year (model input)
brand_key, model_key	Text	Normalised keys used only to join a new-car mid price
new_mid_lakh	Numerical	Matched new-car mid ex-showroom (lakh); blank if no match
residual_pct	Numerical	$\text{asking_price_lakh} / \text{new_mid_lakh} \times 100$; hold-value screen only

Table 1. Columns in the cleaned analysis dataset (6,629 rows). Model inputs are age_years, km_driven, ownership, fuel_type, transmission, body_type, brand and city.

3. Data Preparation and Exploratory Analysis

Cleaning and derived fields

Duplicate advertisements were removed using the unique appointment id. Vehicle age was derived as 2026 minus manufacturing year and restricted to 0–25 years. Kilometres were kept between 200 and 250,000, while asking prices were restricted to ₹0.4–150 lakh to remove scrape errors and extreme luxury outliers. Pickup and truck bodies were excluded so that the study remained focused on passenger cars. Text fields and brand aliases were standardised before analysis.

Where a new-car mid ex-showroom price could be matched on brand and model, residual value is asking price as a percentage of that new mid-price (4,392 of 6,629 listings, 66%). Unmatched rows are left blank; no new price is invented. All 404 Ford listings are unmatched because Ford no longer sells new passenger cars in India. Residual percent is used only as a hold-value screen and is not an input to the prediction model. Remaining missing model inputs are handled inside the pipeline: numeric fields use median imputation and categorical fields use the most frequent category.

Descriptive analysis and visualisations

Asking prices are spread out, not bunched: mean 6.69 lakh, median 5.50 lakh, standard deviation 5.31 lakh (interquartile range 3.43–8.49 lakh). Median age is 7 years and median kilometres 68,431. First-owner cars are 64% of the sample; 69% are manuals. Figure 2 shows the price and age distributions.

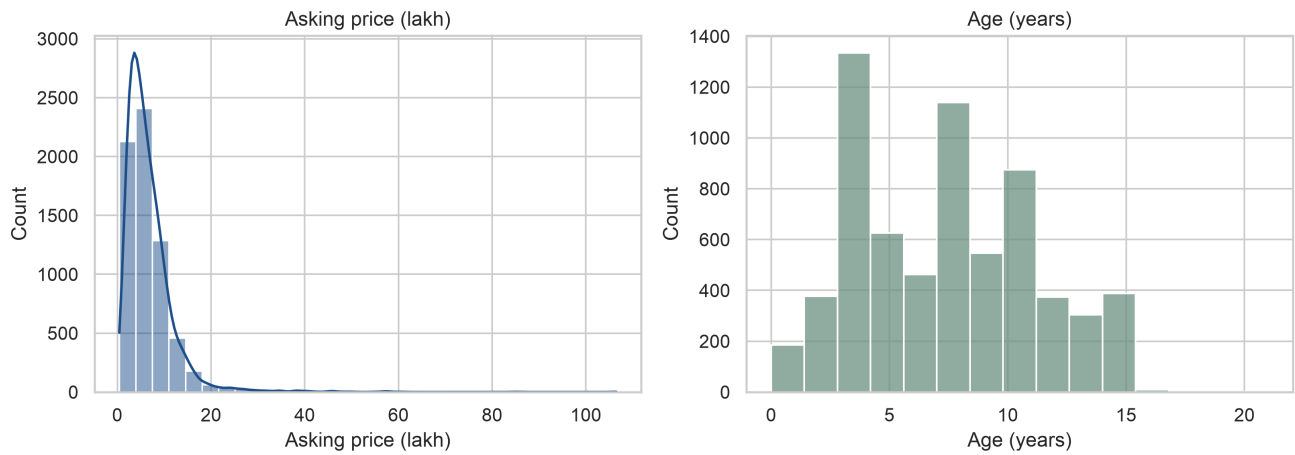


Figure 2. Asking price (lakh) and age (years) in the cleaned sample ($n = 6,629$).

Age has a negative association with asking price ($r = -0.46$). Kilometres show a weaker association in the same direction ($r = -0.23$). The two should be read together: a young high-kilometre listing is not equivalent to an older low-kilometre listing, so a single per-year markdown is a weak pricing rule. Figure 3 is the age–price view; Figure 4 is the kilometre view.

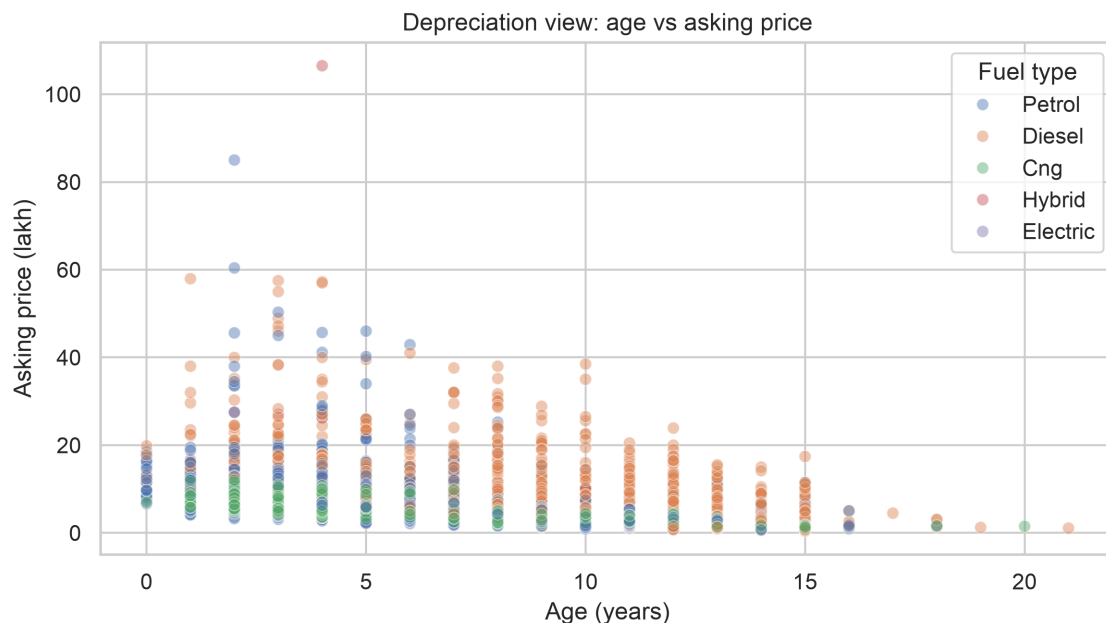


Figure 3. Age versus asking price, coloured by fuel type.

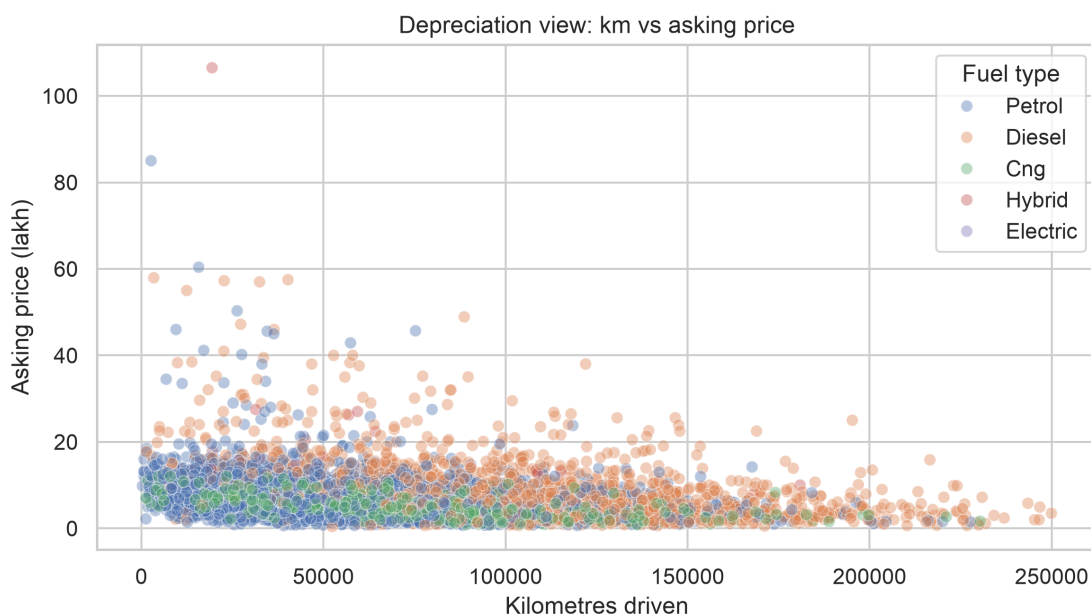


Figure 4. Kilometres driven versus asking price, coloured by fuel type.

Table 2 puts age and kilometres together. A young car with high kilometres still lists around ₹7.96 lakh ($n = 260$), while an older car with low kilometres lists around ₹2.90 lakh ($n = 142$). Age is the stronger observed association of the two: even high-kilometre young cars remain well above older stock. Kilometres still vary within the same age band — young low-km cars sit at ₹8.50 lakh — which again argues against a single yearly markdown.

Age band	Low km ($\leq 50k$)	Mid km (50–90k)	High km ($>90k$)
Young (0-5 yrs)	8.50 ($n=1,449$)	7.70 ($n=813$)	7.96 ($n=260$)
Mid (6-10 yrs)	4.55 ($n=550$)	4.70 ($n=1,167$)	4.99 ($n=921$)
Older (11+ yrs)	2.90 ($n=142$)	2.80 ($n=504$)	2.80 ($n=823$)

Table 2. Median asking price in lakh (and listing count) by age band and kilometre band.

Median asking price is higher for diesel (6.23 lakh) than petrol (5.40 lakh), and higher for SUVs (7.55 lakh) than sedans (4.74 lakh) or hatchbacks (3.25 lakh). That pattern largely reflects what those cars cost when new, not a separate depreciation story. Figure 5 compares fuel and body medians.

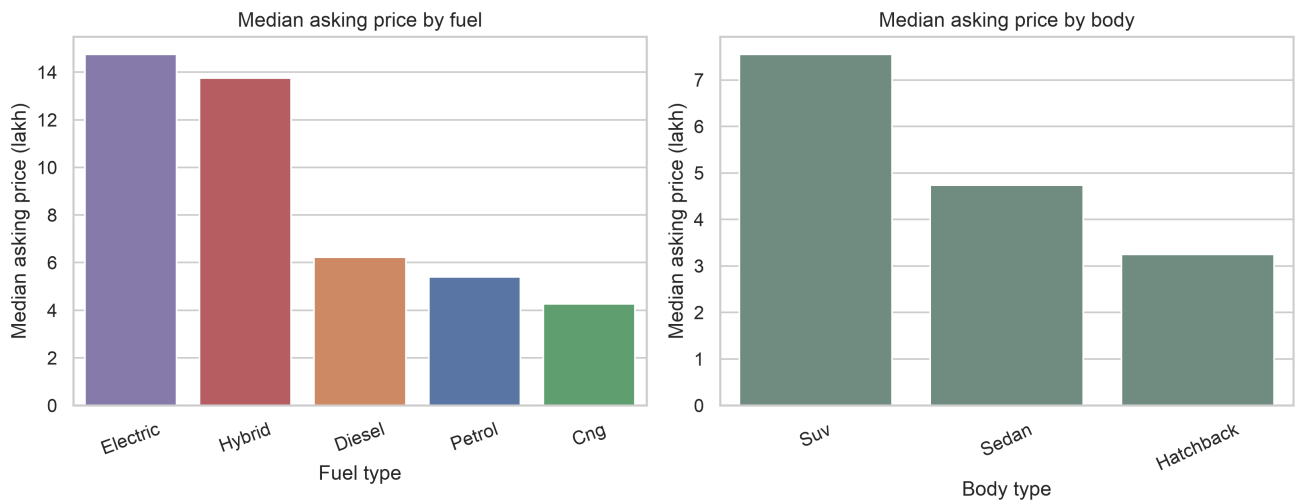


Figure 5. Median asking price by fuel type and body type.

City medians among the 12 markets with the most ads still differ: Hyderabad 6.67 lakh versus Surat 4.31 lakh, with New Delhi near 4.71 lakh. City is included in the forest but ranks below age, brand and kilometres in feature importance. Figure 6 shows those twelve medians.

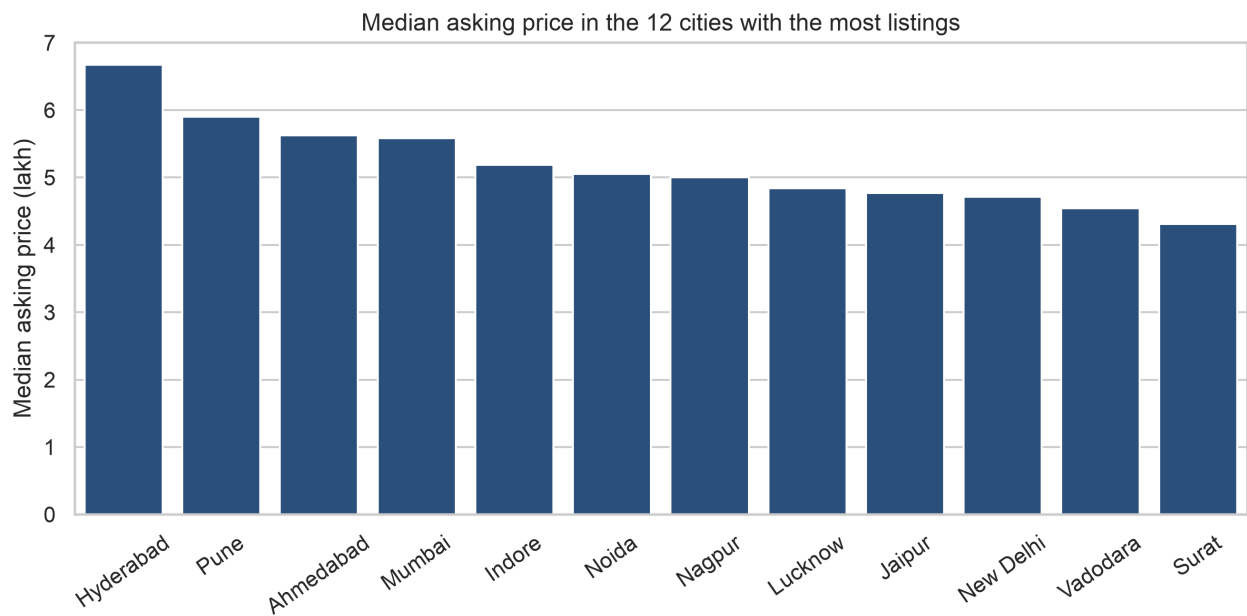


Figure 6. Median asking price in the 12 cities with the most listings.

On the 4,392 listings with a new-car match, median residual value is 58% of the current new mid-price. Kia listings show a higher median residual (83%) than Mercedes (16%), but Mercedes stock in this sample is older and still lists higher in rupees (median ask 10.7 lakh versus Maruti 4.08 lakh). Figure 7 shows residual percent declining with age. Ford used cars remain in the price model; they have no residual percentage because no current new-car match exists. Table 3 reports residual percent by brand together with median age: it is a snapshot of the matched sample, not a same-age brand ranking.

Brand	Matched listings	Median residual %	Median ask (lakh)	Median age (years)
KIA	430	83%	9.11	4
Mahindra	237	83%	11.00	4
Maruti	499	59%	3.71	9
Honda	420	40%	4.40	9
Hyundai	294	38%	6.16	8
BMW	111	18%	13.50	12
Mercedes Benz	88	16%	12.45	11
Audi	88	13%	7.60	13

Table 3. Residual value (asking price as a % of matched new-car mid price) for selected brands.

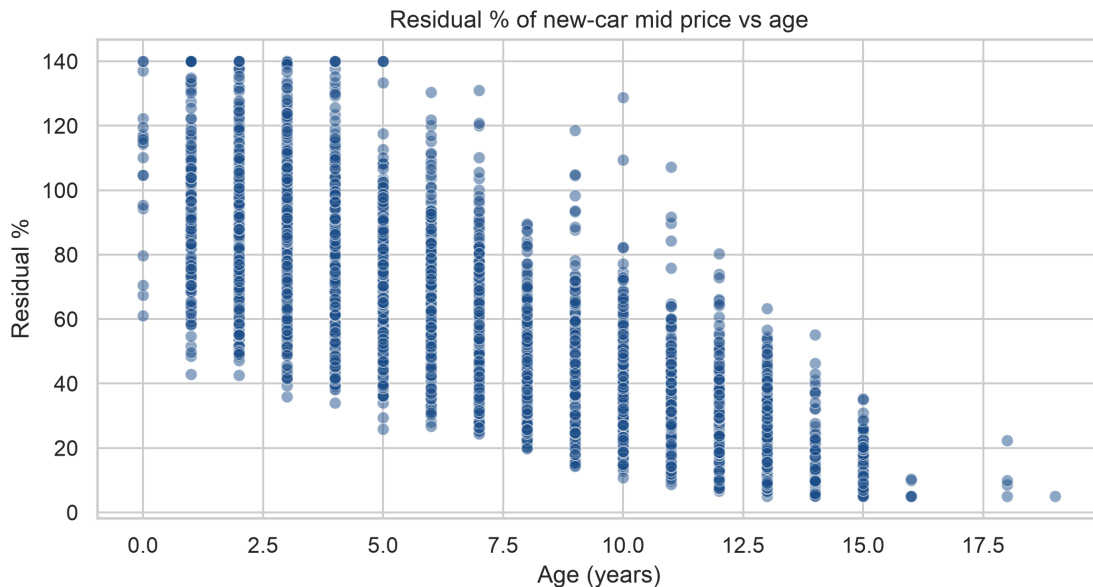


Figure 7. Residual value as a percentage of matched new-car mid price versus age ($n = 4,392$).

4. Analytics Method and Implementation

The task is a supervised regression problem: predict asking price in lakh from listing attributes. Three methods from the Business Analytics syllabus are implemented on the same features and the same 80/20 split (5,303 training and 1,326 test listings), with five-fold cross-validation as a stability check. Predictors are age, kilometres, ownership count, fuel, transmission, body type, brand and city. New-car mid-price and residual percentage are excluded to avoid leakage. Numeric gaps are imputed with the median and scaled; categorical fields are imputed with the most frequent category and one-hot encoded.

Linear Regression is the interpretable baseline: it tests whether the selected attributes have an approximately additive relationship with price. The Decision Tree is included because used-car pricing can change at thresholds—for example after a certain age or kilometre range—and because interactions

between brand, age and transmission may not be linear. Tree depth is limited to reduce overfitting. Random Forest averages 280 trees so that those non-linear patterns are captured more stably than with a single tree. The models are compared using test R^2 , MAE, RMSE and cross-validation R^2 . Performance numbers and business interpretation are reported in Section 6.

Item	Setting
Linear Regression	Baseline; scaled numeric inputs after median imputation
Decision Tree	max_depth = 8; random_state = 42
Random Forest	n_estimators = 280; max_depth = 14; min_samples_leaf = 2; random_state = 42
Train–test split	80% train (5,303), 20% test (1,326)
Cross-validation	5-fold CV R^2 on the training side of the same pipeline
Numeric fields	Median imputation and standard scaling
Categorical fields	Most-frequent imputation and one-hot encoding
Excluded from X	New-car mid-price and residual_pct (leakage)

Table 4. Model configuration used in the notebook.

5. Comparison with State-of-the-Art Methods

Four recent published studies on used-car price prediction are compared with this case study. The comparison is on method, dataset, evaluation approach, findings, strengths and limitations. A higher reported score in another paper is not treated as evidence that that method is better here, because the datasets and experimental settings differ.

Published study / year	Dataset	Method used	Evaluation metric	Key result	Comparison with this work
Cui et al. (2022), Electronics	2021 MathorCup competition data: 30,000 training records, 20 named + 15 anonymous features (China)	Iterative XGBoost + LightGBM; residual-network predictions fused in as new features (DXL)	Blended score $M = 0.2 \times (1 - \text{MAPE}) + 0.8 \times (\text{share of predictions within 5\% relative error})$, reported as “accuracy”	DXL reached $M = 75\%$, above Random Forest (51%), XGBoost (53%), LightGBM (55%) and a residual network alone (63%) on the same file	Uses boosting and deep features beyond the course syllabus, on a Chinese valuation-contest dataset, not real listings. This study stays with Linear Regression, Decision Tree and Random Forest on scraped Indian Cars24 listings and reports rupee MAE/RMSE.

Bhatt et al. (2023), IEEE GCITC	Large public/Kaggle used-car file with many listing attributes	Linear, Ridge, Lasso, Bayesian, Decision Tree, Random Forest, XGBoost, LightGBM, Gradient Boosting, MLP	MAE, MSE, R^2	Broad empirical comparison of linear, tree, boosting and neural regressors	Uses a ready-made dataset and a much wider model list. This study's primary file is a direct Cars24 scrape (6,629 cleaned listings) and is limited to three syllabus models under one split.
Sanke et al. (2024), IJCA	Web-scraped OLX and Cars24 listings (India)	Gradient Boosting, Random Forest, CatBoost; chatbot interface	Hold-out "accuracy" (80/20 split)	CatBoost ~80.03%; Random Forest ~78.63%; Gradient Boosting ~78.05%	Closest data source: they also scrape Cars24. They report accuracy rather than R^2 /MAE, add a chatbot, and do not analyse residual value.
Li (2024), HBEM	>2,000 listings; BMW, Volkswagen, Acura and Tesla	Linear Regression, Decision Tree, Random Forest; K-fold CV	R^2 , MAE, RMSE	Random Forest R^2 0.8562; Decision Tree 0.796; Linear Regression 0.530	Same three model families. The sample is smaller, four brands only, and a different market, so the published R^2 cannot be compared with this study's 0.800.

Table 5. Comparison of this case study with four recent published used-car pricing studies.

On method, Cui et al. [1] and Bhatt et al. [2] go beyond the syllabus (iterative boosting, neural nets, many regressors). Li [4] uses the same three families as this study. Sanke et al. [3] also use tree ensembles, plus a chatbot. This case study keeps Linear Regression, Decision Tree and Random Forest so that the models can be justified and compared under one split. That is a strength for interpretability and a limitation relative to published boosting pipelines.

On dataset, Sanke et al. [3] are closest because they scrape Cars24 (with OLX). Cui, Bhatt and Li use other used-car files, including packaged public data. This study's primary file is a collected Cars24 scrape (6,629 cleaned listings). On evaluation, the papers do not share one metric: Cui reports a blended MAPE/relative-error-accuracy score, Sanke reports hold-out accuracy, and Bhatt and Li report R^2 /error. This study reports test R^2 , MAE, RMSE and cross-validation R^2 on asking price in lakh. Those figures describe this split only. Li's Random Forest R^2 of 0.856 and this study's 0.800 are not ranked against each other. A strength of the present work is residual value and age-kilometre analysis; a limit is that condition and sold price are unobserved, as in most listing studies.

6. Results, Business Insights and Recommendations

Model performance

Model	Test R^2	MAE (lakh)	RMSE (lakh)	5-fold CV R^2
Linear Regression	0.668	1.827	2.914	0.609
Decision Tree	0.716	1.662	2.695	0.664
Random Forest	0.800	1.379	2.262	0.777

Table 6. Test-set and cross-validation performance of the three models trained on the same Cars24 split.

Linear Regression achieves test R^2 0.668. This confirms that the selected attributes contain substantial pricing information, but its MAE of ₹1.827 lakh shows that one linear relationship cannot fully represent brand premiums, age thresholds and feature interactions.

The Decision Tree improves test R^2 to 0.716 and lowers MAE to ₹1.662 lakh. Its gain over the linear baseline demonstrates useful non-linearity. However, its 5-fold cross-validation R^2 of 0.664 (± 0.064) is below its test result, and that fold-to-fold spread is the widest of the three models, so a single tree is the least stable choice under resampling.

Among the three models evaluated on the same test set and under the same experimental setup, Random Forest achieved the highest test R^2 (0.800) and the lowest MAE (₹1.379 lakh) and RMSE (₹2.262 lakh), with cross-validation R^2 0.777 (± 0.040), the closest of the three models to its own test score. It reduces MAE by about 24.5% versus Linear Regression and 17.0% versus the Decision Tree. The typical absolute error is about 25% of the sample median (5.5 lakh). RMSE remains above MAE because expensive outliers produce larger squared errors. Therefore Random Forest was selected as the final model for this case study.

Figure 8 compares Random Forest predictions with actual prices. Listings in the common ₹0–15 lakh range stay closer to the diagonal; errors widen for expensive cars because luxury stock is less common and condition and variant details are not fully observed.

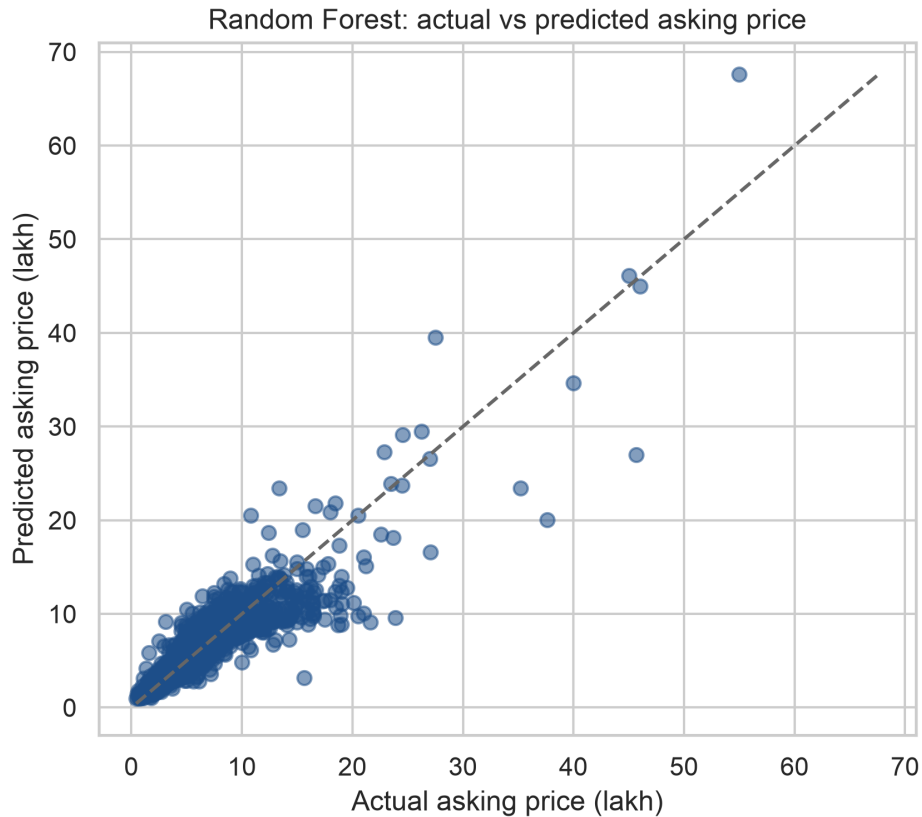


Figure 8. Random Forest: actual versus predicted asking price. The dashed line is a perfect prediction.

Figure 9 isolates the error itself (predicted minus actual) against actual price, which makes the pattern in Figure 8 sharper. Residuals sit in a tight band close to zero for everyday listings, then fan out in both directions once actual price passes about ₹15 lakh. The spread of residuals is about 3.5 times wider above ₹15 lakh (standard deviation 5.42 lakh) than below it (1.56 lakh), while the mean residual overall is essentially zero (−0.06 lakh). This indicates the model is not biased high or low on average, but it is precise only on the common price range and materially less precise on the small luxury tail — the same caution the recommendations below give for luxury listings now has a direct error-based number behind it.

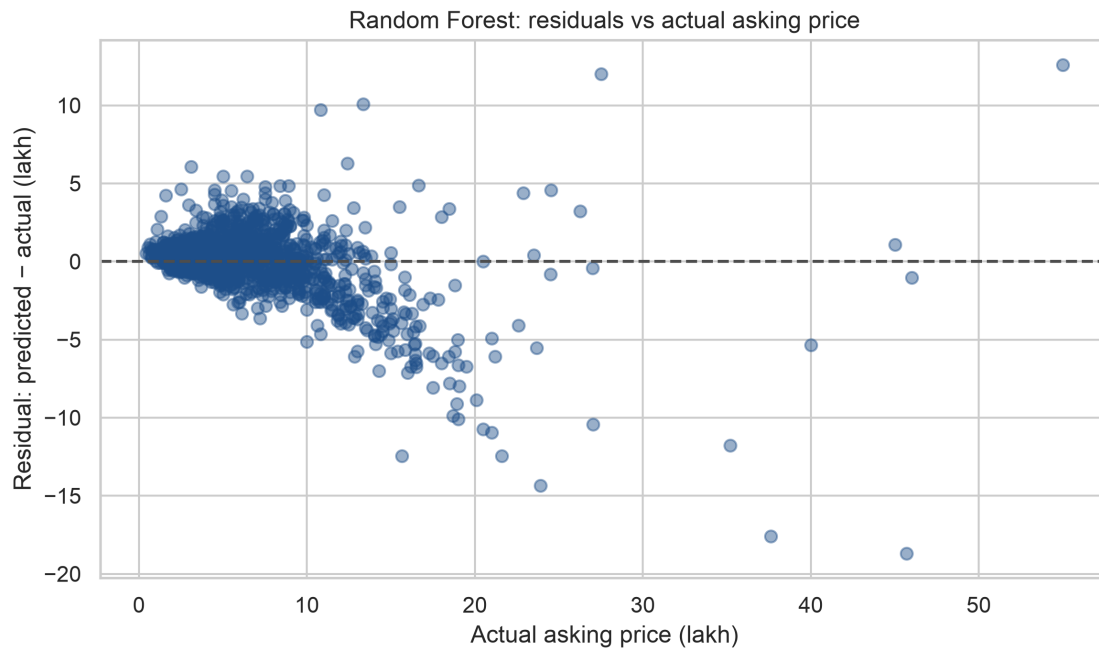


Figure 9. Random Forest residuals (predicted – actual) versus actual asking price.

Figure 10 reports importance after one-hot encoding, so each bar is a column, not an original field. Age is the strongest observed pricing signal. Mercedes and BMW flags mark a higher rupee band (used Mercedes median 10.7 lakh versus Maruti 4.08 lakh); they do not imply slower percentage depreciation (Mercedes residual 16%). Kilometres rank next, then manual transmission and diesel. City is included but is not among the leading columns, which is consistent with the modest city-median gaps in Figure 6.

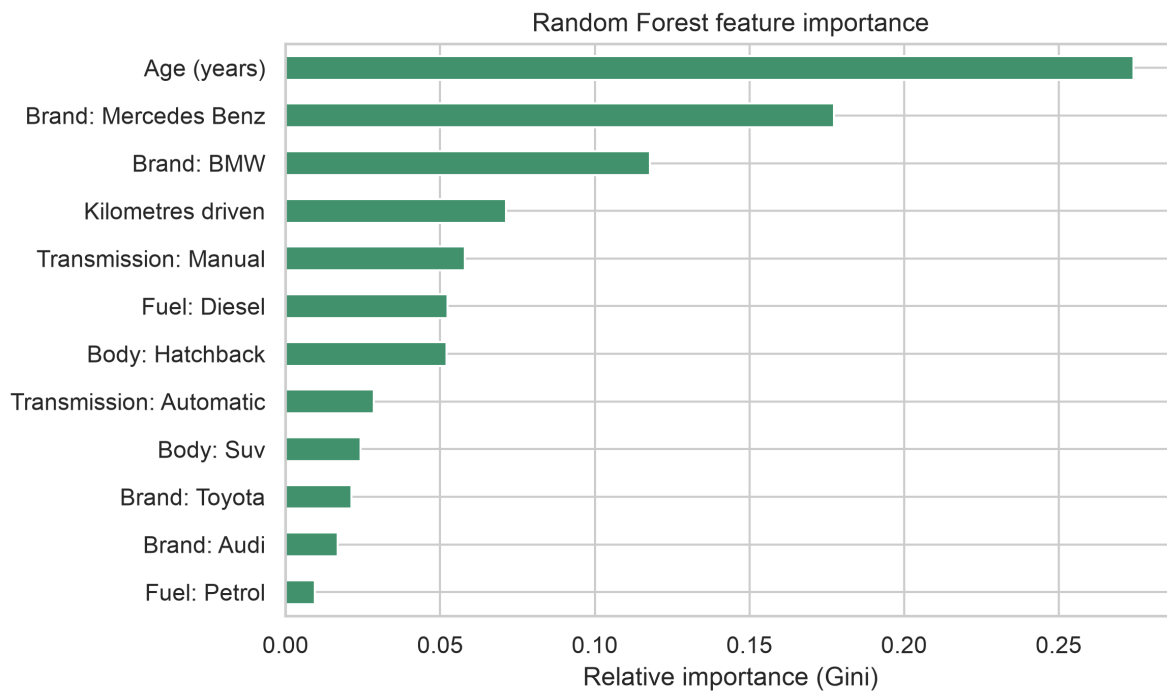


Figure 10. Random Forest feature importance (top 12 columns after encoding).

Business insights and recommendations

Used asking prices are dispersed (median 5.5 lakh, interquartile range about 3.4–8.5 lakh). Age is the strongest observed pricing signal in this dataset ($r = -0.46$). Kilometres show a weaker association with asking price than age ($r = -0.23$). The forest ranks the same order, so a young high-kilometre listing and an older low-kilometre listing should not share one yearly markdown.

Luxury still commands a rupee premium in the second-hand market. A used Mercedes lists around 10.7 lakh against about 4.1 lakh for a Maruti in the same pool. That is absolute value, not hold of today's new price — Mercedes residual is about 16%, while younger volume names such as Kia sit near 83%. Higher-priced luxury listings need separate caution: they are fewer in the sample and prediction errors are larger. A higher rupee ask does not mean slower percentage depreciation.

SUVs (median 7.5 lakh) and diesels (6.2 lakh) list above hatchbacks (3.2 lakh) and petrol (5.4 lakh). That pattern is consistent with higher new-car prices in those segments rather than a separate used-market premium. A typical matched used listing is about 58% of a current new mid-price. High-volume cities are not interchangeable: Hyderabad's median ask is 6.7 lakh against 4.3 lakh in Surat.

For pricing decisions, the Random Forest estimate should be treated as the centre of a negotiation band rather than an automatic final quote. MAE is an average absolute error, not a formal prediction interval. Listings whose asking prices differ substantially from the estimate should be reviewed for factors that are not fully represented in the dataset, such as vehicle condition, service history, accident history, exact variant and documentation. The first adjustment should reflect age and kilometres, followed by brand, body type, fuel, transmission and the local market.

City-specific comparisons should be used instead of copying a national median. Residual percentage should be used for trade-ins only when a credible current new-car match exists; no reference should be

invented for discontinued brands. Luxury predictions also require greater caution because high-ticket listings are fewer and show larger errors.

The results have four important limitations. Asking price is not the final negotiated sale price. Condition, accident history, service history and exact equipment are not fully represented. Residual value uses a current nameplate-level new-car midpoint rather than the original invoice or exact variant price. Finally, the data is a market snapshot, so inventory and prices can change over time.

7. Conclusion

Public listing attributes can support a more defensible used-car asking price than a single nearby advertisement or a flat yearly markdown. Two scrapes of the same Cars24 pages (Python and Apify) both recovered about 6,700 unique listings. The cleaned Python file of 6,629 cars is the analysis dataset. Asking price is dispersed (median 5.5 lakh). Age is the strongest observed pricing signal ($r = -0.46$); kilometres show a weaker association than age ($r = -0.23$). Where a new-car mid price could be matched, median residual value is about 58% and is a hold-value screen, not a model input.

Among Linear Regression, Decision Tree and Random Forest on the same split, Random Forest achieved test R^2 0.800 and MAE ₹1.379 lakh. Scores from other published studies are not ranked against this result. In operations, the estimate is a negotiation centre: screen from age and kilometres, then adjust for brand and the local market, and use residual percent only where a current new-car match exists. Asking price is not a cleared sale, and condition is unobserved, so the model remains a screen rather than an automatic quote.

8. References

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